One Stream, Many Windows: Soft Targets for Consolidating Fast and Slow Conversions

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Abstract. Conversion pipelines in advertising are split by how long a conversion takes to observe: an upper-funnel onsite conversion is visible within minutes of the click, while a down-funnel offsite conversion takes a day or more. Each objective defines an *attribution window*, the elapsed time within which a conversion counts, and conventionally a training pipeline's join delay is set equal to that attribution window. Tying the two together forces every objective onto its own pipeline, each as stale as the window it waited for. We ask whether they can be set independently. Writing training records sooner turns every conversion still in flight into a negative example, and a streaming pipeline that visits each click at most once cannot revise them afterwards. Yet most published corrections rely on exactly such a revision: of 9 methods we reproduce, 7 write more than one training record per click, as many as 4.00. On the Criteo Delayed Feedback Dataset we compare three targets across 5,520 runs, four served capacities, three join delays, two attribution windows and two teacher architectures. Writing the record early and replacing only the negatives with a teacher model's estimate beats waiting for the matured label in 23 of 24 configurations, by 1.90% at seven days and 0.70% at one, which is 76% and 79% of what a label-peeking oracle attains. It is the strongest method that obeys the single-visit constraint.

Keywords: Conversion rate prediction · Delayed feedback · Online advertising · Knowledge distillation · Streaming training pipelines.

1 Introduction

Advertising systems predict several types of conversion at once, and those types differ sharply in how long they take to observe. A website view or an add-to-cart is recorded within minutes of the click; a purchase, an app install or a submitted lead may take a day or a week. Each objective is defined with an *attribution window*, the elapsed time within which a conversion is credited to the click. Conventionally that window also determines the *join delay*, the age at which the click's training record is written to the stream: the pipeline waits until the label is knowable, then emits the record. This convention has two disadvantages. First, a model for a slow objective trains only on data at least an attribution window old, so it never sees fresh data during online training. Second, a system that ranks several objectives jointly must either maintain one pipeline per objective

or delay all of them until the slowest window closes, imposing the staleness of the slowest funnel on the fastest.

Neither disadvantage is intrinsic to conversion rate modelling. While the attribution window is fixed by the business definition of a conversion, the join delay is a pipeline choice. We ask what happens when the two are set independently, and specifically whether an objective with an attribution window of days can be trained from records joined within minutes.

The obstacle is that an early-joined record is wrong: every conversion still in flight when the record is written is recorded as a negative. A substantial literature corrects this censoring bias [1,11,10,2,4], but most corrections revise the record afterwards, by duplicating the click when its outcome lands, by relabelling it, or by adding prediction heads at future horizons. A streaming pipeline that appends each record once and trains on it once can do none of these. We state that restriction as a *single-visit* constraint.

Our idea is that this censoring bias can be mitigated by the prediction of a model that has been trained on the matured outcomes in the conventional way, that is, a *teacher model*. Such a teacher is often readily available in production, where large ranking models are already trained across objectives on matured data [8]. The contributions of this work are: (1) a separation of the join delay from the attribution window, together with the causality and single-visit constraints that a deployable pipeline must satisfy, and an audit of 9 published methods finding that 7 of them violate single-visit by writing more than one record per click; (2) evidence that writing the record early and replacing only its observed negatives with teacher model predictions beats waiting for the matured label; and (3) the finding that at small served capacities the hard label can be dropped altogether, since a training pipeline that uses only the teacher's estimates as labels can beat a conventional pipeline whose join delay equals the attribution window, which benefits early ranking stages such as retrieval, where model complexity is low.

2 Problem Setting and Proposed Approach

A click i occurs at time C_i with features x_i and, if it converts, does so after a delay D_i. The attribution window v fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$. The join delay o fixes when the record is written, so the label available at that moment is

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

Setting $o = v$ reproduces the conventional pipeline. Setting $o \ll v$ is what we want, so that fast and slow conversions can share one training pipeline.

What a deployable pipeline must satisfy. At no moment can we train on conversions that have not yet landed, which is a straightforward **causality rule**. In addition, we have the **single-visit** constraint: let n_i be the number of times a click i enters the served model's loss over its lifetime. This constraint, widely used in production systems, requires $\max_i n_i \le 1$.

Which published methods comply. We reproduced 9 published methods from scratch under a single protocol and counted the records each one's served model consumes per distinct click. Of these, 7 write more than one. At $v = 1\,\mathrm{d}$ the counts are FTP [5] 4.00, MISS [6] 3.05, PI [13] 2.00, DDFM [3] 1.89, DFM [1] 1.33, DEFUSE [2] 1.03, ES-DFM [10] 1.03. Only FSIW and ULC satisfy the single-visit constraint. Of the two, FSIW corrects the loss and leaves the target itself untouched, while ULC changes the training target.

Proposed methods. We study four ways of generating the pipeline's training target. FRESH writes at $\Delta = o$ with $\tilde{y}_i = Y_i^{(o)}$, applying no correction at all. WAIT writes at $\Delta = v$ with $\tilde{y}_i = Y_i^{(v)}$, which is the conventional method and our comparison baseline. CORRECT-DISTILL writes at $\Delta = o$, keeps the observed labels and softens only the observed negatives with a teacher model's prediction.

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \,\Big|\, Y^{(o)}{=}0,\, x\Big)\,. \tag{2}$$

An observed positive stays exactly 1, and only the uncertain mass moves. This is the functional form used by ULC [9]; the two differ in how w^T is estimated. DISTILL-ONLY goes further still, replacing every observed label with the teacher's prediction. The teacher model used by both DISTILL-ONLY and CORRECT-DISTILL is itself trained with the conventional WAIT method. Finally, ORACLE writes at $\Delta = o$ but labels the record $Y_i^{(v)}$. It violates the causality rule and is not deployable. We include it because it bounds what any causal correction can recover.

3 Experiments and Results

3.1 Experimental Setup

Data and protocol. We use the public Criteo conversion logs released with [1]. Clicks span 60 days. Days $[0, 30)$ initialise the model and days $[30, 60)$ form the online training period, divided into 719 hourly rounds yielding 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a seven-point grid on a held-out slice of the initialisation period, then pinned across seeds. The teacher model's rate is fixed and decoupled from that of the serving model.

Model architecture. In the model backbone, the eight integer fields enter a dense arm as $\log(1 + x)$ paired with an observed-mask bit, normalised and passed through a two-layer perceptron to a projection vector. The nine categorical fields are hashed per field and embedded, giving a token per field. Tokens and the dense projection are then consumed by a stack of Wukong interaction layers [12], which produce the final prediction.

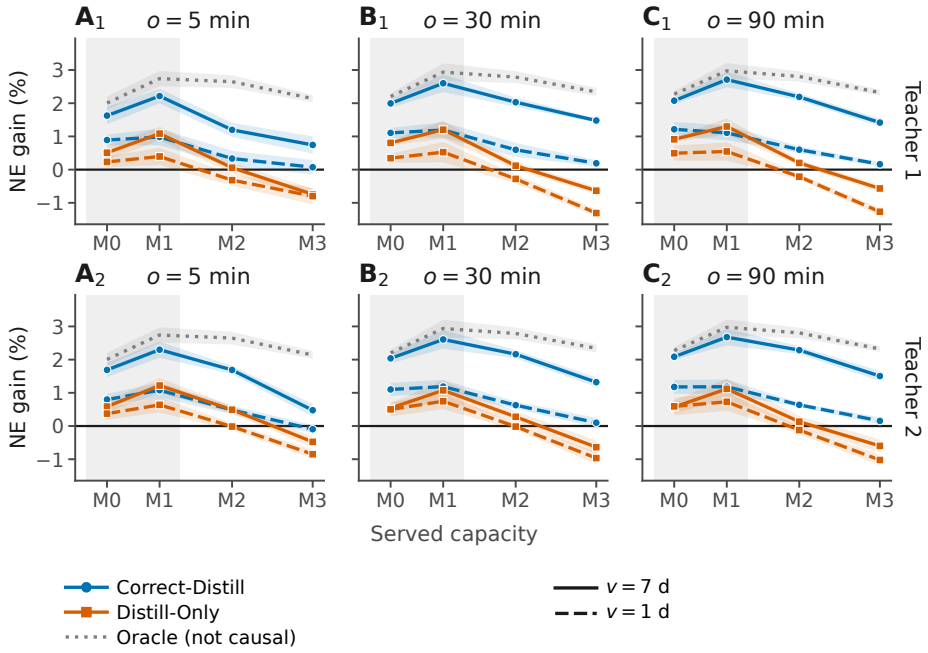


Fig. 1. Relative NE reduction against WAIT as a function of served capacity, for three join delays o and two teacher models. Shading on a curve is the standard deviation of the per-seed paired differences.

Different setups and evaluation metric. Two attribution windows ($v \in \{1, 7\}$ days), three join delays ($o \in \{5, 30, 90\}$ minutes), four serving-model complexities (M_0 to M_3), and two different teacher models are used. Across the four complexities the hash width per categorical field grows $36 \to 256 \to 1{,}280 \to 16{,}384$ and the embedding dimension widens $4 \to 4 \to 8 \to 8$, roughly an order of magnitude in parameters per step, 1,890 to 1,184,439 parameters and 0.001 to 0.0095 MFLOPs per inference. For the two teacher models, *Teacher 1* places 8-dimensional embedding tables under a two-layer perceptron, giving 9,456,897 parameters at 0.039 MFLOPs. *Teacher 2* places 48-dimensional tables under a deep interaction trunk, giving 57,272,515 parameters at 1.593 MFLOPs. Running both lets us report the method's sensitivity to the choice of teacher rather than to a single arbitrary configuration. For each configuration we ran five to ten seeds for statistical analysis, and we report normalised entropy and calibration against the matured $Y^{(v)}$ for all methods.

3.2 Findings

With the teacher's prediction as a soft target, decoupling the join delay from the attribution window improves accuracy. CORRECT-DISTILL, which writes the record early and replaces only its observed negatives with the teacher's estimate,

Table 1. All reproduced methods at $v = 1\,\mathrm{d}$ and $o = 90$ minutes, the operating point closest to a large-scale production setting, at the smallest and largest served capacities. NE and calibration error are absolute and lower is better; ΔNE is the per-seed paired reduction against WAIT in percent, higher is better, reported with the half-width of its 95% t interval. The replicate is a seed, five per arm and ten for WAIT, and a (teacher, seed) pair for the two arms in bold, which are ours and ran under both teachers. ‡ORACLE writes one record but labels it with the matured outcome, so it violates causality and is an upper reference rather than a method.

	M_0 (smallest)			M_3 (largest)		
Method	NE	Cal.	ΔNE (%)	NE	Cal.	ΔNE (%)
Single visited: yes						
WAIT	0.7865	0.0578	ref.	0.7197	0.0743	ref.
ORACLE‡	0.7769	0.0489	$+1.23 \pm 0.28$	0.7169	0.0677	$+0.40 \pm 0.09$
CORRECT-DISTILL	0.7772	0.0555	$+1.19 \pm 0.12$	0.7186	0.0760	$+0.16 \pm 0.05$
ULC [9]	0.7773	0.0529	$+1.18 \pm 0.22$	0.7188	0.0797	$+0.13 \pm 0.11$
FSIW [11]	0.7785	0.0553	$+1.03 \pm 0.26$	0.7193	0.0780	$+0.07 \pm 0.08$
DISTILL-ONLY	0.7821	0.0672	$+0.56 \pm 0.14$	0.7277	0.0744	-1.11 ± 0.10
FRESH	0.7929	0.1855	-0.80 ± 0.25	0.7357	0.1819	-2.20 ± 0.06
Single visited: no						
FTP [5]	0.7786	0.0515	$+1.02 \pm 0.18$	0.7162	0.0567	$+0.50 \pm 0.15$
PI [13]	0.7806	0.0692	$+0.76 \pm 0.43$	0.7177	0.0871	$+0.29 \pm 0.15$
DEFUSE [2]	0.7789	0.0681	$+0.98 \pm 0.30$	0.7201	0.0955	-0.04 ± 0.14
ES-DFM [10]	0.7791	0.0690	$+0.96 \pm 0.31$	0.7203	0.0966	-0.07 ± 0.13
DDFM [3]	0.7902	0.0632	-0.46 ± 0.31	0.7214	0.0562	-0.22 ± 0.20
DFM [1]	0.7569	0.0612	$+3.78 \pm 0.19$	0.7236	0.0578	-0.53 ± 0.04
MISS [6]	0.7830	0.0826	$+0.45 \pm 0.24$	0.7238	0.1292	-0.56 ± 0.20

beats WAIT in 23 of 24 cells, by 1.90% on average at $v = 7\,\mathrm{d}$ and 0.70% at $v = 1\,\mathrm{d}$. Figure 1 shows where that gain comes from and where it runs out. Comparing $v = 7\,\mathrm{d}$ with $v = 1\,\mathrm{d}$, the gain is largest where the funnel is slowest, because WAIT at $v = 7\,\mathrm{d}$ trains on week-old data and no amount of label accuracy compensates for that. Along the model complexity axis, DISTILL-ONLY, which relies purely on the teacher model's prediction and carries no observed label at all, beats WAIT in all 12 of 12 cells at the two smallest capacities, by 0.60% at $v = 1\,\mathrm{d}$ and 0.85% at $v = 7\,\mathrm{d}$, and loses at the two largest by 0.50% and 0.14%. This trend is plausibly because a more complex model learns better from hard targets, so the freshness gained from a soft target no longer compensates for the information lost in replacing a hard target with a soft one [7]. The fully soft target nonetheless remains applicable to early ranking stages such as retrieval, where model complexity must stay low in order to rank a large number of items at high throughput. Finally, comparing across join delays $o = 5$, 30 and 90 minutes, a larger o improves the accuracy of CORRECT-DISTILL at $v = 7\,\mathrm{d}$, likely because a larger o admits more hard labels.

Table 1 places both targets against the literature at one operating point, $v = 1\,\mathrm{d}$ and $o = 90$ minutes. Among the methods a single-visit pipeline can actually run we lead at both rungs, +1.19% against +1.18% for ULC [9] at M_0 and +0.16% against +0.13% at M_3, with overlapping intervals, and our calibration error tracks WAIT's at both rungs while FRESH degrades it by 3.2 and 2.4 times. The constraint is not free: DFM reaches +3.78% at M_0 by writing the same click several times, which supplies training signal that no single-visit method can recover.

4 Conclusion and Future Work

An objective's attribution window is fixed by the business definition of a conversion; the join delay its training pipeline adopts is not. We have shown that the two can be set independently. Writing the record within minutes and replacing only its observed negatives with a teacher's estimate beats waiting for the matured label in 23 of 24 configurations and recovers 76% of what a label-peeking oracle attains at seven days, and it does so under the single-visit constraint that most published corrections violate. The correction is cheap to adopt, being insensitive to the teacher's architecture.

The next step is consolidating several ranking objectives into one model and one stream, an active industrial direction [8] whose first obstacle is one of pipelines rather than of modelling. Under WAIT, objective k's record exists only at $C_i + v_k$, so a shared append-only stream must either write K records per click, violating single-visit K times over, or delay every objective until $\max_k v_k$ and impose the slowest funnel's staleness on the fastest. The target studied here removes that dilemma in principle: a single stream at one short join delay can carry $\tilde{y}_{i,k} = Y^{(o)}_{i,k} + (1 - Y^{(o)}_{i,k})\, w_k^T(x_i)$ for every objective at once. In future work, we will generalise this to K prediction heads, K attribution windows, one stream, and a production-scale teacher model.

References

1. Chapelle, O.: Modeling delayed feedback in display advertising. In: Proc. KDD. pp. 1097–1105 (2014)
2. Chen, Y., Jin, J., Zhao, H., Wang, P., Liu, G., Xu, J., Zheng, B.: Asymptotically unbiased estimation for delayed feedback modeling via label correction. In: Proc. WWW. pp. 369–379 (2022)
3. Dai, S., Zhou, Y., Xu, J., Wen, J.R.: Dually enhanced delayed feedback modeling for streaming conversion rate prediction. In: Proc. CIKM. pp. 390–399 (2023)
4. Ding, C., Wu, J., Yuan, Y., Li, C., Wang, X., Wang, D., Yang, F., Rabinovich, A.: Delayed feedback modeling with influence functions. Proc. AAAI **40**(17), 14648–14656 (2026)
5. Li, H., Pan, F., Ao, X., Yang, Z., Lu, M., Pan, J., Liu, D., Xiao, L., He, Q.: Follow the prophet: Accurate online conversion rate prediction in the face of delayed feedback. In: Proc. SIGIR. pp. 1915–1919 (2021)

6. Liu, Q., Ao, X., Guo, Y., He, Q.: Online conversion rate prediction via multi-interval screening and synthesizing under delayed feedback. Proc. AAAI **38**(8), 8796–8804 (2024)
7. Menon, et al.: A statistical perspective on distillation. In: Proc. ICML. PMLR, vol. 139 (2021)
8. Meta AI: Sequence learning: A paradigm shift for personalized ads recommendations. Meta AI Engineering Blog (2025), `https://ai.meta.com/blog/generative-ads-recommendation-model-gem/`
9. Wang, Y., Sun, P., Zhang, M., Jia, Q., Li, J., Ma, S.: Unbiased delayed feedback label correction for conversion rate prediction. In: Proc. KDD. pp. 2456–2466 (2023)
10. Yang, J.Q., Li, X., Han, S., Zhuang, T., Zhan, D.C., Zeng, X., Tong, B.: Capturing delayed feedback in conversion rate prediction via elapsed-time sampling. Proc. AAAI **35**(5), 4582–4589 (2021)
11. Yasui, S., Morishita, G., Fujita, K., Shibata, M.: A feedback shift correction in predicting conversion rates under delayed feedback. In: Proc. WWW. pp. 2740–2746 (2020)
12. Zhang, B., Luo, L., Chen, Y., Nie, J., Liu, X., Guo, D., Zhao, Y., Li, S., Hao, Y., Yao, Y., Lakshminarayanan, G., Wen, E.D., Park, J., Naumov, M., Chen, W.: Wukong: Towards a scaling law for large-scale recommendation (2024)
13. Zhang, X., Li, W., Li, R., Fu, Z., Tang, T., Zhang, Z., Chen, W.Y., Noorshams, N., Jasapara, N., Ding, X., Wen, E., Feng, X.: Personalized interpolation: Achieving efficient conversion estimation with flexible optimization windows. In: Proc. CIKM (2025)